MILITARY SPECIFICATION

STARTER, FIRE, NP3, M2

This Specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirement and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of fire starter designated as Fire Starter, NP3, M2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, Flat, on Seals
PPP-B-26	- Bag, Plastic, Polyethylene
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Box, Fiberboard
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-650 - Explosive: Sampling, Inspection and Testing

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

B4-1-15 - Chamber, Fuel, Assembly
C4-1-14 - Starter, Fire, NP3, M2, Assembly
D4-1-31 - Palletization for Starter, Fire, NP3, M2
D4-1-32 - Marking Packing (Starter, Fire, NP3, M2)
C4-2-70 - Unit Packaging, Fire Starter (For 5 Fire Starters)
C4-2-121 - Unit Packaging, Fire Starter (For 1 Fire Starter)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing C4-1-14 and associated drawings and with all retirements specified in applicable specifications and standards.

3.3 Leakage. The fuel chamber assembly, Drawing B4-1-15, shall not leak when tested as specified in 4.5.2.

3.4 Functioning/waterproofness. The fire starter shall function satisfactorily as evidented by the igniting and complete burning of all combustible components when tested as specified in 4.5.3. Burning time shall be 3 minutes minimum. There shall be no exploders.

3.5 Compliance of chemical mixtures. For each chemical mixture, the contractor shall Provide a certificate of analysis to these requirements, as determined by the Contracting Officer.

3.6 First article inspection. This specification contains technical provisions for First Article Inspection. Requirements for the submission of First Article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free from burrs, chips, sharp-edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. Exterior surface coating shall be continuous except for a few light scratches not exposinq base material. The cleaning method used shall not be injurious to any of the parts nor shall the parts by contaminated. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of-this detail specification.

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4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a First Article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The First Article sample shall consist of the following items in sample quantities as indicated,

Part Description	Drawing	Quantity
Fire Starter, NP3, M2	C4-1-14	30
Fuel Chamber Assembly	B4-1-15	30
Fuel Chamber-pellet Assembly	B4-1-16	30
Case Assembly	B4-1-17	30
Pellet, Match Head	A4-1-18	30
Cover, Fuel Chamber	A4-1-19	30
Cover, Ignition Chamber	A4-1-22	30
Wire, Scratch (Coated)	B4-1-23	30
Wire, Scratch	C4-1-24	30
Handle, Scratch Wire	B4-1-26	30

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-13175H(AR)

PARAGRAPH	TITLE: Starter, Fire, NP3, M2		SHEET 1 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case Assembly (Dwg. B4-1-17)				
	Examination for defects	30		3.2	4.4.2.1
	Cover, Fuel Chamber (Dwg. A4-1-19)				
	Examination for defects	30		3.2	4.4.2.2
	Cover, Ignition Chamber (Dwg. A4-1-22)				
	Examination for defects	30		3.2	4.4.2.3
	Pellet, Match Head (Dwg. A4-1-18)				
	Examination for defects	30		3.2	4.4.2.4
	Volatile content	5 gr.		3.2	(a)
	Wire, Scratch (Uncoated) (Dwg. C4-1-24)				
	Examination for defects	30		3.2	4.4.2.5
	Curvature of wire to coiled end incorrect	(b)		3.2	Gage

NOTES:
(a) Determine per MIL-STD-650 para. 101.5 or 101.6.
(b) Same items subject to examination for defects.

DSMC-DA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-13175H(AR)

PARAGRAPH	TITLE: Starter, Fire, NP3, M2		SHEET 2 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Wire, Scratch (Coated) (Dwg. D4-1-23)				
	Examination for defects	30		3.2	4.4.2.6
	Handle, Scatch Wire (Dwg. B4-1-26)				
	Examination for defects	30		3.2	4.4.2.7
	Fuel Chamber Assembly (Dwg. B4-1-15)				
	Examination for defects	30		3.2	4.4.2.8
	Fuel Chamber-Pellet Assembly (Dwg. B4-1-16)				
	Examination for defects	30		3.2	4.4.2.9
	Fire Starter, NP3, M2 (Dwg. C4-1-14)				
	Examination for defects	30		3.2	4.4.2.10
	Functioning/Waterproofness	(a)		3.4	4.5.3

NOTES:
(a) Samples shall be taken from the 30 above.

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4.4 Quality conformance inspection.

4.4.1. Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of fire starters shall contain:

a. Cellulose nitrate material bearing the same lot interfix number from one manufacturer.

b. Parts bearing the same lot number from one manufacturer.

c. Match Head Mixture-VI from one batch or blending or batches which results in homogeneity of all Match Head Mixture-VI used in loading the entire fire starter lot.

d. Scratcher Mixture-I from one batch or blending or batches which results in homogeneity of all Scratcher Mixture-I used in loading the entire fire starter lot.

e. Incendiary Mixture NP3 from one batch or blending or batches which results in homogeneity of all Incendiary Mixture NP3 used in loading the entire fire starter lot.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and Tests tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Case, Assembly and Detail				B4-1-17
					NEXT HIGHER ASSEMBLY
					B4-1-15
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Body length incorrect		0.40%	3.2	Gage
102	Distance of partition from end of ignition chamber incorrect		0.40%	3.2	Gage
103	Partition not securely sealed to body		0.40%	3.2	Visual
Minor					
201	Thickness of partition incorrect	(a) (b)	0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:
(a) Thickness measurement taken before assembly.
(b) Partition may be treated as stamped part according to para. 4.5.4.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Cover, Fuel Chamber				A4-1-19
					NEXT HIGHER ASSEMBLY
					B4-1-15
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height incorrect	(a)	0.40%	3.2	Gage
102	Width incorrect	(a)	0.40%	3.2	Gage
Minor					
201	Thickness incorrect		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

(a) Cover may be treated as stamped part according to para. 4.5.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Cover, Ignition Chamber				A4-1-22
					NEXT HIGHER ASSEMBLY
					C4-1-14
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height incorrect	(a)	0.40%	3.2	Gage
102	Width incorrect	(a)	0.40%	3.2	Gage
Minor					
201	Thickness incorrect		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES: (a) Cover may be treated as stamped part according to para. 4.5.4.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Pellet, Match Head				A4-1-18
					NEXT HIGHER ASSEMBLY B4-1-16
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pellet height incorrect		0.40%	3.2	Gage
102	Position of hole incorrect. (If drilled before assembly)		0.40%	3.2	Gage
103	Moisture and volatile content greater than .5% of dry weight		0.40%	3.2	(a)
Minor					
201	Pellet bottom length incorrect		0.65%	3.2	Gage
202	Pellet bottom width incorrect		0.65%	3.2	Gage
203	Pellet top length incorrect		0.65%	3.2	Gage
204	Pellet top width incorrect		0.65%	3.2	Gage
205	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

(a) Determine per MIL-STD-650, Method 101.5 or 101.6.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Wire, Scratch (Uncoated)				C4-1-24
					NEXT HIGHER ASSEMBLY
					B4-1-23
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Wire length incorrect		0.65%	3.2	Gage
202	Diameter of coil incorrect		0.65%	3.2	Gage
203	Curvature of wire to coiled end incorrect		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.00%	3.7	Visual
NOTES:					

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Wire, Scratch (Coated)				B4-1-23
					NEXT HIGHER ASSEMBLY
					C4-1-14
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Scratch mixture extends along pull wire beyond max.		0.40%	3.2	Gage
102	Diameter of scratch mixture end incorrect		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Handle, Scratch Wire				B4-1-26
					NEXT HIGHER ASSEMBLY C4-1-14
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	Metal thickness		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Chamber, Fuel, Assembly				B4-1-15
					NEXT HIGHER ASSEMBLY
					B4-1-16
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Leak Test		100%	3.3	4.5.2
Major					
101	Void space in fuel chamber beyond maximum limit		0.40%	3.2	Gage
102	Weight of Incendiary Mixture NP3 less than 3.9 grams		0.40%	3.2	Gage
103	Fuel chamber cover not securely sealed to body		0.40%	3.2	Visual
104	Alternate end closure A4-1-19 not flush to .010 below body end		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:

NPSMC-NA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Fuel Chamber-Pellet Assembly				B4-1-16
					NEXT HIGHER ASSEMBLY
					C4-1-14
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pellet not cemented securely to fuel chamber		0.40%	3.2	Visual
102	Position of hole thru pellet incorrect		0.40%	3.2	Gage
103	Diameter of hole thru pellet incorrect		0.40%	3.2	Gage
Minor					
201	Position of hole above pellet incorrect		0.65%	3.2	Gage
202	Diameter of hole above pellet incorrect		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Fire Starter, NP3, M2				C4-1-14
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Scratcher mixture in contact with match head pellet (Before cover secured to ignition chamber end)		100%	3.2	Visual
Major					
101	Any component missing, damaged, or incorrectly assembled		0.40%	3.2	Visual
102	Labels missing, incorrect, or damaged		0.40%	3.2	Visual
103	Functioning/Waterproofness		0.40%	3.4	4.5.3
104	Wire breakage during functioning	(a)	0.40%	3.4	4.5.3.1
Minor					
201	Cover not properly secured to ignition chamber end		0.65%	3.2	Visual
202	Ignition chamber end not properly painted		0.65%	3.2	Visual
203	Scratch wire not properly secured to case		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES: (a) Same samples used in functioning test above.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Unit Packaging-Fire Starter (For one Fire Starter)				C42-121
					NEXT HIGHER ASSEMBLY
					PL4-2-121
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box broken or torn		0.40%	3.2	Visual
Minor					
201	Marking incorrect or missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

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MIL-S-13175H(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Unit Packaging - Fire Starter (For 5 Fire Starters)	1 OF 1	C4-2-70
			NEXT HIGHER ASSEMBLY: PL-4-2-70

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box broken or torn		0.40%	3.2	Visual
102	Less than 5 fire starters in package		0.40%	3.2	Visual
103	Filler missing or incorrectly placed		0.40%	3.2	Visual
Minor					
201	Marking incorrect or missing		0.65%	3.2	Visual
202	Box not correctly closed		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Intermediate Packaging	1 OF 1	NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box broken or torn		0.40%	3.2	Visual
102	Less than 100 fire starters per box		0.40%	3.2	Visual
103	Polyethylene bag torn or improperly sealed (Level A only)		0.40%	3.2	Visual
Minor					
201	Box not correctly sealed		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Packing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged		0.40%	3.2	Visual
102	Less than 500 fire starters per box		0.40%	3.2	Visual
103	Incorrect style, class or grade of box		0.40%	3.2	Visual
Minor					
201	Box not correctly closed		0.65%	3.2	Visual
202	Strapping or tape (Level C only) missing or not properly applied		0.65%	3.2	Visual
203	Contents not packed tightly		0.65%	3.2	Visual
204	Filler pads not positioned around all sides, top and bottom (Level A and B only)		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER D4-1-32
4.4.2.15	Marking, Packing (Before Closing)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any marking, incorrect or illegible		0.40%	3.2	Visual
102	Labels (when used) not overcoated with water-resistant adhesive		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13175H(AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
4.4.2.16	Palletization for Fire Starter				D4-1-31
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Palletization procedure incorrect		0.40%	3.2	Visual
102	Strapping missing, loose or incorrectly applied		0.40%	3.2	Visual
103	Proper number of boxes incorrect		0.40%	3.2	Visual
104	Label not overcoated with water-resistant adhesive		0.40%	3.2	Visual
Minor					
201	Labels incorrect or missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirement of Section 3 of this specification shall be ascertained by-current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Leakage. The fuel chamber assembly, Dwg. B4-1-15, shall be stored for a period of 4 ± 1/2 hours in an oven at a temperature of 155° ± 5°F. Evidence of liquid fuel on the exterior surface shall constitute leakage. (See 6.5)

4.5.3 Functioning/waterproofness. The fire starter shall be placed horizontally in water at a temperature of 75° ± 5°F to a depth of 6 inches measured from the top surface of the starter for a period of 4 + 1/2 hours. At the end of the soak period, the fire starter shall be removed from the water and wiped dry. Within one hour after removal from the water the fire starter shall be functioned as specified on the label (Dwg. D4-1-27). During functioning, the fire starter should be protected from drafts.

4.5.3.1 Wire breakage. If the wire breaks during functioning the fire starter may be functioned by either pulling the wire with pliers or by igniting with a wooden match. This defect will be reported separately from other functioning failures and evaluated independently.

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4.5.4 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamped process for use in production. In additon, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer.

5. PACKAGING

5.1 Packaging. The packaging shall be level A or C as specified in the contract or order.

5.1.1 Level A.

5.1.1.1 Type I unit packaging. The fire starter shall be packaged one per package in accordance with Dwg. C4-2-121.

5.1.1.2 Type II unit packaging. The fire starter shall be unit packaged five per package in accordance with Dwg. C4-2-70.

5.1.1.3 Intermediate packaging. The fire starter, unit packaged in accordance with either type I or type II above, as specified in the contract or order, shall be intermediate packaged, 100 fire starters per box in fiberboard boxes conforming to style RSC, Grade W6c or W5c of PPP-B-636, having inside dimensions in inches of 7-1/4 by 6-1/2 by 3-1/2, length, width, and depth, respectively. Unit packages shall be arranged on end, ten rows of ten each for type I and two rows of ten each for type II. The box shall be closed method IV in the appendix to PPP-B-636 and sealed within a polyethylene bag conforming to type II, style 1 of PPP-B-26, having dimensions in inches of 11-1/4 by 13. Air volume in the sealed bag shall be kept to a practicable minimum.

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5. 1.2 Level C. Fire starters shall be packaged in accordance with 5.1.1 except that intermediate package fiberboard box may be domestic class and the polyethylene bag shall be omitted.

5.2 Packing. The packing shall be level A, B, or C as specified in the contract or order.

5.2.1 Level A. The fire starters packaged in accordance with 5.1.1, shall be packed, 500 per box in wood boxes conforming to style 4, Class 2, Grade A, 0-50 pound limit for type 1 or 2 load of PPP-B-621, having inside dimensions in inches of 20-1/2 by 8 by 7-1/4 length, width, and depth, respectively. Filler pads fabricated of either W6c or W5c fiberboard, PPP-F-320, shall be included one per face at all interior faces of the box to separate the packages from the box. Additional pads shall be included as necessary to restrict free movement of the packages. The boxes shall be closed and strapped in accordance with the appendix to PPP-B-621. Strapping shall conform to either Class I or IV, Grade A or B of QQ-S-781.

5.2.2 Level B. The fire starters shall be packed in accordance with 5.2.1 except the wood box shall be Grade B, non-preservative treated.

5.2.3 Level C. The fire starters, packaged in accordance with 5.1.2, shall be packed, 500 per box in fiberboard boxes having inside dimensions in inches of 20-1/2 by 7-3/4 by 7-1/4 length, width, and depth, respectively, and conforming to style RSC, Grade V3c of PPP-B-636. The boxes shall be closed in accordance with method III in the appendix to PPP-B-636 and reinforced by banding with either tape or non-metallic strapping as specified therein.

5.3 Palletization. When specified in the contract or order, packed fire starters shall be palletized in accordance with Dwg. D4-1-31 for full pallet load quantities. Adjustments shall be made accordingly for less than full pallet load quantities if feasible for palletization.

5.4 Marking. In conjunction with information furnished in the contract or order, marking shall be as specified herein and in the Code of Federal. Regulations, Title 49 - Transportation, Parts 100-199.

5.4.1 Unit package marking. Unit packages shall be marked in accordance with Dwg. C4-2-121 or Type I and Dwg. C4-2-70 for Type II, as appropriate. l

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5.4.2 Intermediate Package marking. The fiberboard box of the intermediate package shall be marked on one side with identification marking consisting of national stock number, nomenclature (item description) quantity and unit of issue, and level of packaging and date (month and year) packaged. Marking material shall be in accordance with D4-1-32. Marking shall be minimum 12 point type size and shall be applied prior to sealing within the polyethylene bag when applicable.

5.4.3 Packing marking. The shipping container shall be marked in accordance with Dwg. D4-1-32.

5.4.4 Palletized load marking. palletized unit loads shall be marked in accordance with Dwg. D4-1-31, adjusting quantity as applicable if less than full pallet load.

6. NOTES

6.1 Intended use. The fire starter is intended to start fires under adverse weather conditions.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to: Commander, USAAMCCOM, ATTN: DRSMC-QAT-I (D), Dover, NJ 07801.

6.4 Precautions. Because of the hazardous nature of filling munitions with chemical agents, filling installations are advised to take all precautions to prevent fires and to protect personnel from injury during the preparation of incendiary mixtures.

6.5. Leakage. Leakage may be determined more readily when the fuel chambers are placed with adequate spacing on a layer of absorbent paper (blotting paper or paper towel) and gently weighted to hold them in contact with the absorbing paper.

6.6 Drawing. Drawings listed in Section 2 of this specification under the heading “US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-13175H	STARTER, FIRE, NP3, M2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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MILITARY SPECIFICATION

STARTER,FIRE,NP3,M2

This Amendment forms a part of Military Specification MIL-S-13175H(AR), dated 15 May 1984, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.5: Compliance of Chemical Mixture.

Delete In its entirety and substitute: " For each chemical mixture an analysis shall be performed as specified in para. 4.5.5."

PAGE 25

Add the following new paragraph:

4.5.5, Chemical Analysis.

4.5.5.1, Scratcher Mixture." The following procedure is to be employed in analyzing the mixture Phosphorous,Red (MIL-P-211, Class C) Sand-Polystyrene. This prodecure is to be applied before solubilizing the binder during processing. Constituent materials are to be in compliance with the applicable specification and drawing. Alternate procedures for this analysis are acceptable only if approved by AMSMC-QAT-I, Dover, N.J. 07801-5001.

1. Polystyrene. Transfer an accurately weighed portion of approximate y two grams of the sample to a tared 30 ml. medium porosity fritted class thimble crucible. Using Soxhlet or Roweg-extraction apparatus and toluene as the solvent, separate the Polystyrene from the other constituents.

FSC 1370

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Completeness of extraction will be indicated by the absence of Polystyrene pellets and should be substantiated by constant values obtained for the weight of the thimble crucible, and contents as the extraction procedure continues. For this purpose remove the thimble crucible from the extraction apparatus and dry it in an oven for 45 minutes at 80 ± 5° C.

$$\% \text{ Polystyrene} = \frac{A-B}{W} \times 100$$

Where: A = Weight of sample and crucible before extraction

B = Weight of sample and crucible after extraction

W = Weight of sample

2. Sand. Transfer the contents of the thimble crucible to a 400 ml beaker using approximately 10 ml. of water. Add 35 ml. of concentrated nitric acid. Cover the beaker with a watch glass, warm on a hot plate until vigorous reaction stops and then boil down to a volume of approximately 20 ml. Add 10 ml. of concentrated perchloric acid and boil down to a volume of 3 or 4 ml. After cooling, transfer the residue in the beaker to a tared 30 ml. medium porosity fritted glass crucible. Using small portions of water to effect a quantitative removal, a wash the contents of the crucible with at least ten 5 ml. portions of water, at room temperature using a glass stirring rod to assure good contact with all portions of the residue. Aspirate between additions of the wash water and finally aspirate to dryness. Place crucible, with residue, in a drying oven maintained at 110 ± 5° C. for a minimum of sixty minutes. Cool the crucible,and residue in a desiccator and obtain its weight.

$$\% \text{ Sand} = \frac{C-D}{W} \times 100$$

Where: C = Weight of residue and crucible

D = Weight of crucible

w = Weight of sample, from Section 1.

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3. Phosphorus Red.

% Phosphorus, Red = 100 - (E+F)

Where: E = % Polystyrene

F = % Sand

4.5.5.2, Match Head Mixture. The following procedure is to be employed in analyzing the mixture Potassium Chlorate (MIL-P-150, Grade B, Class 2), Antimony Sulfide (MIL-A-159, Class 1), Polystyrene. This procedure is to be applied before solubilizing the binder during processing, Constituent materials are to be in compliance with the applicable specification and drawing. Alternative procedures for this analysis are acceptable only if approved by AMSMC-QAT-I,Dover N.J. 07801-5001.

1. Potassium Chlorate. Transfer on accurately weighed portion of approximately one gram of the sample to a tared 30 ml. medium porosity fritted glass crucible. To the material in the crucible, add five 10 ml. portions of water at room temperature. Maintain each portion of the water in contact with the sample for approximately three minutes using a glass stirring rod for mixing. Aspirate after each addition of water. Place the crucible, with residue, in a drying oven maintained at 110 ± 5°C. for a minimum of sixty minutes. Cool the crucible, with residue, in a desiccator and obtain its weight.

$$\% \text{ Potassium Chlorate} = \frac{A-B}{W} \times 100$$

Where: A = Weight of sample and crucible before extraction.

B = Weight of residue and crucible after extraction.

W = Weight of sample

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2. Antimony Sulfide. To the crucible containing the residue, obtained as indicated in Section 1, add six 15 ml. portions of concentrated hydrochloric acid at approximately 80°C. Maintain each portion of acid in contact with the contents of the crucible for approximately three minutes using a glass stirring rod for mixing. Aspirate after each addition of acid. The disappearance of all dark material indicates complete reaction of the antimony sulfide. Wash the contents of the crucible with four 5 ml. portions of 2% hydrochloric acid and finally with four 5 ml. portions of water, both at room temperature. Aspirate thoroughly and place crucible, with contents, in a drying oven maintained at 110 ± 5°C. for a minimum of 60 minutes. Cool the crucible, and residue, in a desiccator and obtain its weight.

$$\% \text{ Antimony Sulfide} = \frac{B-C}{W} \times 100$$

Where: B = Weight of residue and crucible, from Section 1.

c = Weight of residue and crucible after acid extraction

w = Weight of sample, from Section 1.

3. Polystyrene.

% Polystyrene = 100 - (D+E)

Where: D = % Potassium Chlorate

E = % Antimony Sulfide."

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Preparing activity:
Army - AR

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